Classwise Prototype-Guided Alignment Network for Cross-Scene Hyperspectral Image Classification

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***Abstract*— In the past few years, there has been significant progress in hyperspectral image classification (HSIC). However, when the trained classifier on the source scene is directly applied to a new scene, the classification performance tends to dramatically decrease because of the spectral shift phenomenon. Most existing techniques use feature alignment to learn knowledge from labeled scenes to unlabeled scenes, often overlooking the impact of noisy samples and outliers. To tackle this issue, the classwise prototype-guided alignment network (CPGAN) is proposed for cross-scene HSIC. The core idea is that classwise prototypes across scenes are employed as alignment intermediaries to guide cross-scene feature alignment. Specifically, first, spectral–spatial features from different scenes are extracted with a common feature extractor. Then, an uncertainty-aware pseudolabel selection (UPS) is designed to obtain high-confidence pseudolabels for unlabeled target scenes. Finally, a novel classwise prototype-guided alignment method is proposed to simultaneously achieve interdomain and intradomain alignment (IntraDA). The experimental results conducted on three datasets show that our method achieves superior performance compared to other cutting-edge classification algorithms.**

***Index Terms*— Domain adaptation (DA), feature alignment, hyperspectral image classification (HSIC), prototype calibration (ProC), uncertainty estimations.**

I. Introduction

HYPERSPECTRAL imaging can capture both spatial structural information and spectral data across hundreds of contiguous bands. It has emerged as a cutting-edge technology for remote sensing community, finding widespread applications across various fields such as military reconnaissance [1], [2], [3], smart agriculture [4], [5], [6], and marine monitoring [7], [8], [9]. Hyperspectral image classification (HSIC) stands as a pivotal endeavor within hyperspectral image processing. The goal of HSIC is to determine the class of each pixel by analyzing spatial and spectral information. Over the past few decades, HSIC has garnered significant attention.

In recent times, deep learning has demonstrated remarkable promise across diverse domains, owing to its robust capability for nonlinear representation [10], [11], [12], [13], [14]. Thus, many deep neural networks, i.e., convolutional neural network (CNN) [15], [16], [17], [18], [19], [20], [21], graph convolutional network (GCN) [22], [23], [24], [25], [26], and generative adversarial network (GAN) [27], [28], [29], [30], [31], [32], are introduced into HSIC. For example, Chen et al. [33] introduced the CNN to extract important information from HSI for the first time. Hamida et al. [34] designed a 3-DCNN to simultaneously capture spatial and spectral information. Xie et al. [35] greatly improved HSIC accuracy by constructing a spectral–spatial attention module to enhance feature discriminability. Although CNN has demonstrated impressive capabilities in HSIC, it can only handle data with regular structures and cannot process data with non-Euclidean structures. Hence, GCNs have been introduced to deal with irregular structural data. For example, Wan et al. [23] proposed a GCN to leverage multiscale spectral–spatial similarities, thereby enhancing the classification performance at class boundaries. Yu et al. [36] developed a GCN with two branches that merge the benefits of GCN and CNN to improve the classification accuracy while mitigating the oversmoothing issue caused by the deeper network structure. Those aforementioned methods can achieve remarkable classification performance when the training and testing scenes adhere to the assumption of distribution consistency [37], [38], [39], [40]. Unfortunately, it is difficult to maintain this assumption in the real world, since various factors, such as lighting, atmosphere, and imaging sensors, lead to spectral shift.

To overcome the spectral shift issue, many hyperspectral cross-scene classification techniques have been developed [41], [42], [43], in which only label a portion of source data for model training while the unlabeled target sample are used for testing. Most of the existing unsupervised domain adaptation (UDA) algorithms narrow the distribution gap by aligning the statistical distributions across scenes. For example, Zhou and Prasad [44] introduced the convolutional recurrent neural network to obtain important information, followed

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by reducing the feature distance by transforming the mapping features into common subspaces. Qin et al. [45] projected original features from different scenes into an invariant tensor subspace, and then, the subspace alignment scheme is applied to reduce the distribution discrepancy. In addition, several two-stage alignment strategies are proposed to establish a discriminative classification boundary. Liu et al. [46] developed an adversarial adaptation network to extract high-quality features while employing the maximum mean discrepancy (MMD) to align features across different scenes. Li et al. [47] proposed an adaptation network based on contrastive learning to improve the cross-scene HSIC performance, which reduces the distance between samples belonging to the same class while simultaneously increasing the separation between samples from different classes.

In the computer vision field, many UDA methods have emerged, broadly categorized into two types: statistical distribution alignment and self-training alignment. The statistical distribution alignment-based methods typically focus on designing an appropriate distance metric to minimize domain discrepancies. The widely utilized measures include MMD, correlation alignment (CORAL) [48], Wasserstein distance [49], and Kullback–Leibler (KL) divergence [50]. For instance, Rozantsev et al. [51] introduced a two-stream domain adaptation (DA) architecture that applied the MMD to encourage feature consistency, thereby reducing the distribution discrepancy across scenes. Meng et al. [52] minimized the KL divergence between probability distributions across scenes to improve the transferability of the classification approach. The self-training alignment-based methods aim to design an alternative training strategy. First, pseudolabels are generated for the target scene, following which both the pseudolabel of the target scene and source labels are jointly utilized for model retraining. For example, Zou et al. [53] introduced a self-training framework with label regularization and model regularization, which utilized the maximum value of softmax output to gauge the confidence.

These alignment methods have shown significant success in image classification. However, there are still two challenges in HSIC. On the one hand, obtaining high-confidence pseudolabels is crucial for enhancing the performance of UDA methods. However, existing methods often focus on improving prediction scores for true classes while neglecting the suppression of scores for incorrect classes in estimating pseudolabels. This oversight leads to uncertainty propagation issues, promoting error accumulation and diminishing adaptive performance. On the other hand, most of the existing methods compute the classwise prototype by averaging the features corresponding to each class, which makes it difficult to mitigate the impact of noisy samples. Intuitively, noisy samples can lead to the generated class prototype instability, potentially causing some prototypes to deviate from their true representations.

To tackle these issues, a classwise prototype-guided alignment network (CPGAN) is proposed for the cross-scene HSIC. First, a novel classwise prototype-guided alignment is proposed to narrow the cross-scene gap, which is essential to improve the stability and effectiveness of our approach. This module is constituted by interdomain alignment (InterDA) and intradomain alignment (IntraDA). For the InterDA, the distance between prototypes of the same class across scenes is reduced to promote prototype feature consistency. For the IntraDA, the feature of each scene is encouraged to approach its classwise prototype, which makes the distribution of intraclass features more compact. Second, a classification constraint is utilized to update the prototype, which further enhance its discriminability. Finally, due to the lack of a labeled target sample, an effective uncertainty-aware pseudolabel selection (UPS) module is developed to filter the unreliable labels in the target scene and reduce the model's instability caused by noisy pseudolabels. Consequently, the proposed CPGAN significantly reduces the cross-scene discrepancy to achieve superior classification performance. Extensive experiments on three public datasets show that the CPGAN can outperform the competitive approaches. The main contributions of this work are given as follows.

1) A CPGAN is proposed to enhance intraclass consistency and interclass diversity among land covers across different scenes, which is useful to mitigate negative transfer and reduce the domain discrepancy across scenes.
2) A classification constraint-based prototype calibration (ProC) module is proposed to refine classwise prototypes. As far as we know, this is the inaugural endeavor in cross-scene HSIC to employ ProC, enhancing the representational capacity of prototype features for precise classification.
3) The uncertainty-aware pseudolabels' selection is implemented to produce reliable pseudolabels for model training. This approach not only enables the model to grasp essential information from the target scene but also supplies high-quality pseudolabels for the prototype alignment module.

II. Methodology

In this section, the CPGAN is illustrated in detail. First, we introduced the definition of unsupervised cross-scene HSIC. Then, the overall framework of the CPGAN is described. Next, the UPS is presented. Finally, the classwise prototype-guided alignment is elaborated in detail.

A. Problem Definition

Unsupervised cross-scene HSIC aims to transfer knowledge learned from the labeled source scene to the unlabeled target scene. However, due to the spectral shift, it is challenging to achieve satisfactory classification results when directly applying the source classifier to target scene classification tasks. The spectral shift can be categorized into four types, such as covariate shift, conditional shift, label shift, and concept shift. In this work, our primary emphasis is on tackling the covariate shift issue, aiming to align the marginal distribution across different scenes. In specific, selecting n_s labeled source samples and n_t unlabeled target samples to train the classification model, the corresponding constraint based on the prior knowledge is designed to optimize the network and update weights, thereby enhancing the transferability of the

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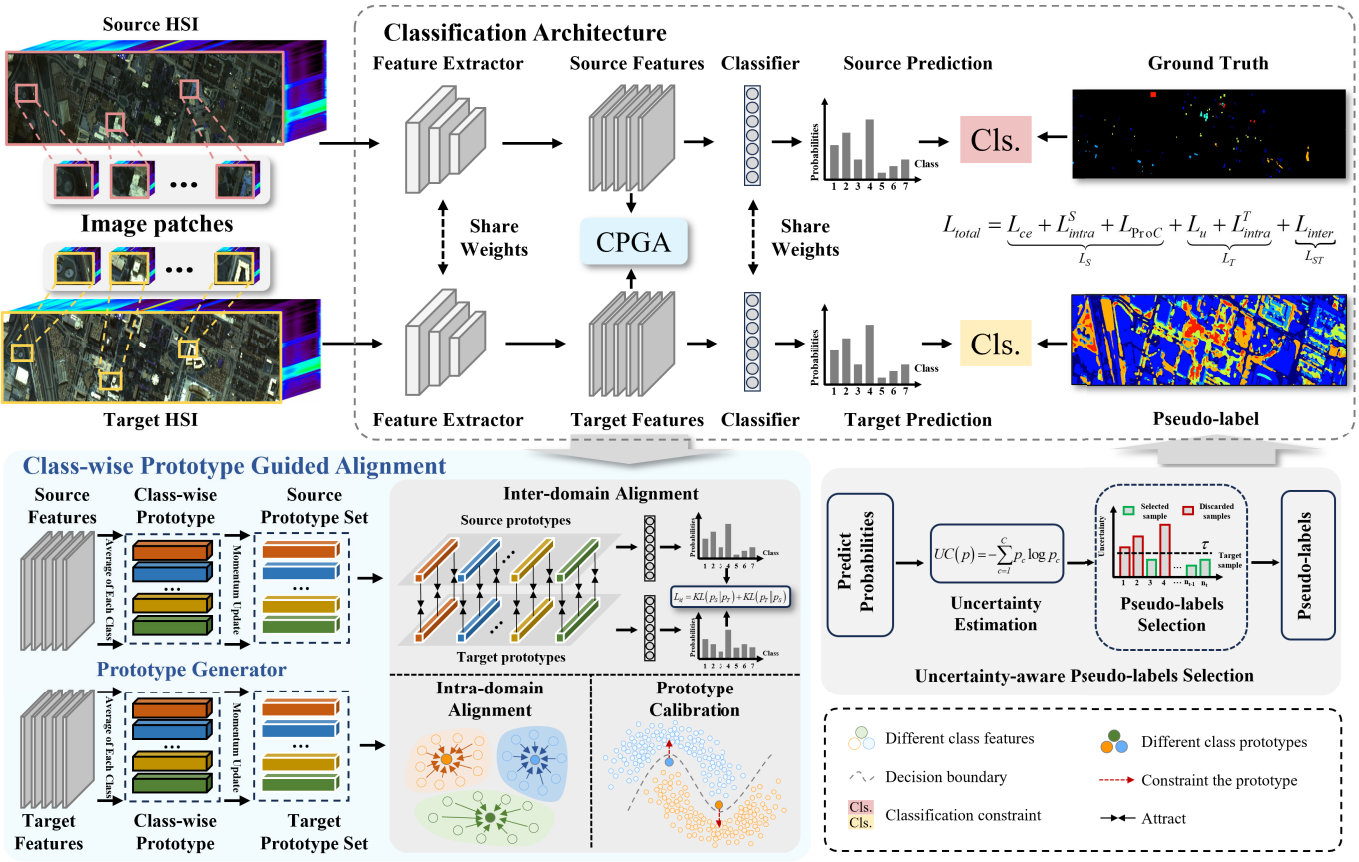


Fig. 1. Flowchart of the proposed CPGAN.

classification model by reducing the distribution discrepancy across scenes. The common constraint L_{com} can be defined as

$$L_{\text{com}} = L_S + L_T + L_{\text{ST}} \tag{1}$$

where L_S represents the source scene loss that is utilized to learn the source scene information. L_T is the target scene loss that aims to extract important information from the target scene, and L_{ST} denotes the interdomain loss that makes the classification model learn interdomain information.

B. Overall Framework

The flowchart of the CPGAN, as depicted in Fig. 1, comprises three key components: supervised learning-based source information learning, classwise prototype-guided alignment, and UPS. Specifically, the training set is first constructed, which contains the labeled source samples $\{\mathbf{X}_S, \mathbf{Y}_S\} = \{\mathbf{x}_s^i, y_s^i\}_{i=1}^{n_s}$ and unlabeled target samples $\{\mathbf{X}_T\} = \{\mathbf{x}_t^i\}_{i=1}^{n_t}$. First, the labeled source samples and the unlabeled target samples are fed into the classification architecture, and supervised learning-based source information learning is utilized to learn the knowledge from the source scene. Second, to classify the target scene, the classwise prototype-guided alignment is adopted to align feature distribution across scenes, which is efficient in improving the transferability of the classification model. Finally, to generate reliable pseudolabels, a UPS is introduced to filter the noisy pseudolabels. Therefore, through the aforementioned techniques, the distribution discrepancy across scenes can be reduced, enhancing the model's classification accuracy in the target scene.

C. Supervised Learning-Based Source Information Learning

Cross-scene HSIC aims to leverage the knowledge learned from a labeled source scene to classify the unlabeled target scene, so it is necessary to accurately learn the source information. In this work, two HSI patches $x_s^i \in R^{r\times r\times B}$ and $x_t^i \in R^{r\times r\times B}$ are separately input into the feature extractor f at first, where r denotes the patch size and B represents the number of spectral bands. The discriminative features across scenes are extracted, $\mathbf{F}_s^i = f(\mathbf{x}_s^i)$, $\mathbf{F}_t^i = f(\mathbf{x}_t^i)$. Then, the classifier FC processes the source feature $\mathbf{F}_s^i$ and the target feature $\mathbf{F}_t^i$ to derive the predicted probabilities $p_s^i = \text{FC}(\mathbf{F}_s^i)$ and $p_t^i = \text{FC}(\mathbf{F}_t^i)$. To learn source information, a widely utilized supervised classification constraint is implemented to optimize the proposed CPGAN, which is calculated as

$$L_{\text{ce}}\left(p_s^c, y_s^c\right) = -\frac{1}{C}\sum_{c=0}^{C}\left(y_s^c \log\left(p_s^c\right) + \left(1-y_s^c\right)\log\left(1-p_s^c\right)\right) \tag{2}$$

where C is the overall count of classes and c denotes the cth class.

D. Classwise Prototype-Guided Alignment

Cross-scene alignment is an effective way to enhance the transferability of the classification model. However, when the

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extracted features in different scenes are directly aligned one by one, it is easy to be affected by noisy samples and outliers, leading to negative alignment and reducing the classification effect. The classwise prototype is defined as the most representative feature within a group of instances belonging to the same class; it exhibits insensitivity to a small number of outliers [54]. Therefore, this article proposes a classwise prototype-guided alignment (CPGA), which enhances the similarity of features within the same category while enlarging the difference between different categories across scenes.

Specifically, we apply the feature extractor f to extract discriminative features across scene images at first. Then, the classwise prototypes are updated based on the minibatch features. For the scene $D \in \{D_S, D_T\}$, D_S and D_T represent the source and target datasets; respectively, we define the prototype of the cth class as $\mathbf{P}_D^c$, which can be calculated as

$$\mathbf{P}_D^c = \frac{1}{\left|\Lambda_D^c\right|} \sum_{(\mathbf{x}_i, y_i) \in \Lambda_D^c} f(\mathbf{x}_i) \tag{3}$$

where Λ_D^c is a training dataset that belongs to the cth category in the minibatch of scene D, i.e., $\Lambda_D^c = \{(\mathbf{x}_i, y_i) \in D | y_i = c\}$.

To make the training phase more robust and less sensitive to individual samples of each minibatch, the exponential moving average (EMA) is applied for classwise prototype update, which can be calculated as

$$\mathbf{P}_D^c|_I = \alpha\, \mathbf{P}_D^c|_I + (1-\alpha)\, \mathbf{P}_D^c|_{I-1} \tag{4}$$

where α represents the exponential decay rate and I is the current iteration.

Based on the classwise prototypes, we focus on uncovering the intrinsic structure of the target scene and strive to acquire more condensed features that facilitate the alignment across scenes. To achieve this objective, we pursue the prototype alignment on the interdomain level of the deep feature space while enforcing intradomain constraints on the feature compactness around diverse prototypes. This ensures the overall consistency of cross-scene distributions, improving the classification accuracy of different scenes.

For the InterDA, we aim to minimize the mean square error (MSE) among prototypes of the same class from different scenes to ensure consistency in their representation, which can be calculated as

$$L_{\text{gp}} = \frac{1}{C} \sum_{c=1}^{C} \left(\mathbf{P}_S^c - \mathbf{P}_T^c\right)^2 \tag{5}$$

where L_{gp} is the MSE value of the cross-scene prototypes. This equation compels the network to provide consistent feature representations for prototypes of the same class across different scenes, thereby enabling accurate transfer of cross-scene knowledge to improve classification accuracy in the target scene.

To further align the interdomain distribution of prototypes across scenes, we apply the KL divergence to ensure consistent assignments between the classification probabilities of source and target prototypes. This can be calculated as

$$L_{\text{kl}} = \frac{1}{C} \sum_{c=1}^{C} \text{KL}\left(\text{FC}\left(\mathbf{P}_S^c\right) \| \text{FC}\left(\mathbf{P}_T^c\right)\right). \tag{6}$$

This equation compels the network to provide consistent prototype labels for adjacent feature points, thereby compelling the model to coherent embedding space aligning the across-scene distributions. The InterDA loss L_{inter} can be calculate as

$$L_{\text{inter}} = L_{\text{gp}} + L_{\text{kl}}. \tag{7}$$

For the IntraDA, the cross-scene samples are embedded in a compact manner around their respective prototypes, so as to facilitate more refined category-level alignment. The following constraint function is utilized for intraclass feature compactness, which can be calculated as follows:

$$L_{\text{intra}}^S = \frac{1}{N} \sum_{c=1}^{C} \sum_{(\mathbf{x}_S^i, y_S^i) \in \mathbf{X}_S^c} \left\| f\left(\mathbf{x}_S^i\right) - \mathbf{P}_S^c \right\|_2^2 \tag{8}$$

$$L_{\text{intra}}^T = \frac{1}{N} \sum_{c=1}^{C} \sum_{(\mathbf{x}_T^i, y_T^i) \in \mathbf{X}_T^c} \left\| f\left(\mathbf{x}_T^i\right) - \mathbf{P}_T^c \right\|_2^2 \tag{9}$$

where L_{intra}^S and L_{intra}^T are IntraDA loss for the source and target scenes, respectively, N represents the number of samples used to calculate the loss during one iteration, and $\mathbf{X}_S^c$ and $\mathbf{X}_T^c$ are defined as a set of samples categorized under class c from the source and target scenes, respectively.

In addition, to enhance the discrimination of prototypes, a ProC constraint is introduced, which ensures that the prototypes can accurately characterize class features, thereby further enhancing the model's generalization ability

$$L_{\text{ProC}} = \frac{1}{C} \left(\sum_{i=1}^{C} L_{\text{ce}}\left(\text{FC}\left(\mathbf{P}_S^i\right), i\right) \right). \tag{10}$$

E. Uncertainty-Aware Pseudolabel Selection

To select high-confidence pseudolabels from the target domain and decrease negative transfer caused by noisy pseudolabels, most of the methods are based on the maximum value of softmax probabilities while ignoring the uncertainty of the classification model. In this situation, it is challenging to accurately choose the high-confidence pseudolabels. To alleviate this issue, the UPS is proposed. Specifically, we estimate the uncertainty of choosing highly reliable pseudolabels, which is calculated as

$$\text{UC}(p) = -\sum_{c=1}^{C} p^c \log p^c \tag{11}$$

where p^c represents the probabilities of the cth class. After obtaining the uncertainty of each pseudolabel, a confidence threshold τ is set to select reliable pseudolabels. Pseudolabels that have high uncertainty are filtered out. Subsequently, the uncertainty mask $\mathbf{M}_u$ can be obtained

$$\mathbf{M}_u = \begin{cases} 1, & \text{UC}(p_t) < \tau \\ 0, & \text{UC}(p_t) > \tau \end{cases} \tag{12}$$

where $\text{UC}(p_t)$ is the uncertainty of the target sample. According to M_u, the reliable pseudolabels are selected from the noisy labels. The corresponding target samples can be utilized to

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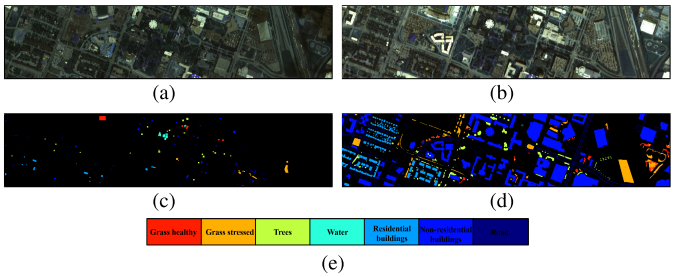


Fig. 2. Houston dataset. (a) False-color image of Houston2013. (b) False–color image of Houston2018. (c) GT image of Houston2013. (d) GT image of Houston2018. (e) Class Name.

update the classification model weights, which are calculated as

$$L_u = -\frac{M_u}{n_t}\sum_{i=1}^{n_t} p_t^i \log p_t^i \tag{13}$$

where L_u represents the constraint to learn target scene information.

Accordingly, by incorporating uncertainty-aware pseudolabels' selection and classwise prototype-guided alignment, the training loss L_{total} of the classification model can be calculated as

$$L_{\text{total}} = \underbrace{L_{\text{ce}} + L_{\text{intra}}^{S} + L_{\text{ProC}}}_{L_S} + \underbrace{L_u + L_{\text{intra}}^{T}}_{L_T} + \underbrace{L_{\text{inter}}}_{L_{\text{ST}}}. \tag{14}$$

III. Experimental Results

To verify the effectiveness of the CPGAN, three widely utilized public hyperspectral datasets are adopted, i.e., the Houston dataset, the Hyrank dataset, and the Shanghai and Hangzhou (S-H) dataset. Seven cutting-edge cross-scene classification algorithms are selected as competitors, including joint adaptation network (JAN) [55], margin disparity discrepancy (MDD) [56], dynamic adversarial adaptation network (DAAN) [57], minimum class confusion (MCC) [58], batch nuclear-norm maximization (BNM) [59], deep subdomain adaptation network (DSAN) [60], and topological structure and semantic information transfer network (TSTnet) [61]. The experimental results are evaluated with three objective indexes, i.e., overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

A. Experiment Datasets

1) Houston: The Houston dataset comprises Houston 2013 and 2018 datasets, which are obtained from different imaging located in the University of Houston campus. The Houston 2013 dataset contains 144 spectral bands with spatial size 349 × 1905 pixels. The wavelength ranges from 380 to 1050 nm. The Houston 2018 dataset has 48 spectral bands with a spatial resolution of 1 m. The spectral range is from 380 to 1050 nm. The two datasets have three common land covers. The common region 209 × 955 is selected for the classification experiment. Fig. 2 presents the false-color images and ground-truth (GT) images. Table I shows the amount of labeled data.

TABLE I
Amount of Labeled Data on the Houston Dataset

Houston2013 and Houston2018 Data			
Index	Class Name	Houston2013 (Source)	Houston2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

TABLE II
Amount of Labeled Data on the Hyrank Dataset

Dioni and Loukia Data			
Index	Class Name	Dioni (Source)	Loukia (Target)
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sderophyllous vegetation	5035	2996
8	Sparce sderophyllous vegetation	6374	2361
9	Sparcely vegetated sand	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
	Total	20024	10317

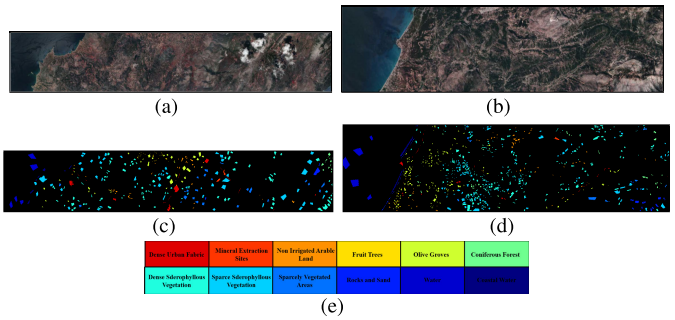


Fig. 3. Hyrank dataset. (a) False-color image of Dioni. (b) False-color image of Loukia. (c) GT image of Dioni. (d) GT image of Loukia. (e) Class Name.

2) Hyrank: The Hyrank dataset is shared by the International Society for Photogrammetry and Remote Sensing (ISPRS) scientific initiatives, which is collected by the Hyperion sensor. This dataset includes two images, i.e., Dioni and Loukia. The two images have 12 common classes. Their spatial sizes are 250 × 1376 and 249 × 945 pixels, respectively. Fig. 3 depicts the false-color images and GT images. Table II shows the amount of labeled data.

3) S-H: The S-H datasets were obtained by the EO-1 sensor. They contain 198 spectral bands. The Shanghai dataset has 1600 × 230 pixels, while the Hangzhou dataset has 590 × 230 pixels. They have three common land covers. Fig. 4 presents the false-color images and GT images. Table III shows the amount of labeled data.

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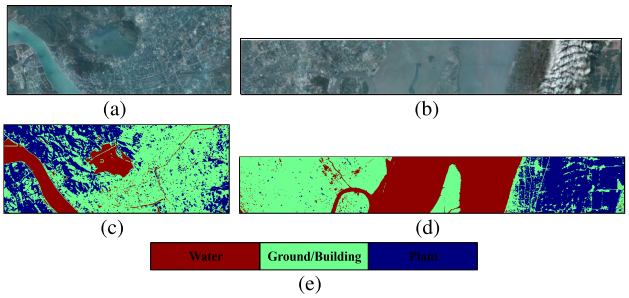


Fig. 4. S-H dataset. (a) False-color image of Hangzhou. (b) False-color image of Shanghai. (c) GT image of Hangzhou. (d) GT image of Shanghai. (e) Class Name.

TABLE III
AMOUNT OF LABELED DATA ON THE S-H DATASET

Hangzhou and Shanghai Data			
Index	Class Name	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
	total	135700	368000

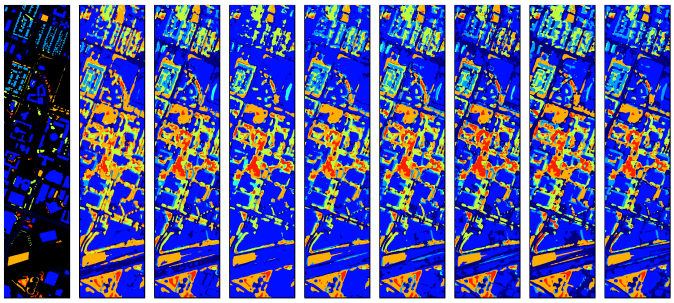


Fig. 5. Classification maps of the Houston task. (a) GT. (b) JAN, OA = 67.98%. (b) MDD, OA = 67.28%. (d) DAAN, OA = 70.91%. (e) MCC, OA = 76.23%. (f) BNM, OA = 70.59%. (g) DSAN, OA = 70.68%. (h) TSTnet, OA = 77.09%. (i) Our method, OA = 80.12%.

B. Parameter Setting

In this work, we conduct all experiments on a server equipped with an NVIDIA GeForce RTX 2080 Ti as the hardware environment. The software configuration is implemented on the PyTorch platform using Python version 3.7.10. We set the value of the input image patch to 13 and the batch size to 100 for all three datasets. The learning process is terminated after 500 epochs. We set the learning rate to 0.5 for the Houston dataset and the Hyrank dataset, while for the S-H dataset, it is set to 0.2. In addition, the stochastic gradient descent (SGD) optimization algorithm is employed to optimize our network. All the presented results are the outcomes of ten random trials for each configuration.

C. Results and Analysis

1) Houston: The first experiment is conducted on the Houston dataset. The size of the training set is set to 5%. Fig. 5 presents the classification maps of all methods. The JAN method fails to identify the "residential buildings" class. The DAAN method produces a similar visual map, which cannot well detect the "residential buildings" class. The JAN method obtains an unsatisfactory classification effect for the "non-residential buildings" class. The MCC method yields noisy labels around the boundaries between diverse land covers. The BNM method enhances the visual impact, but the "grass stressed" and "water" classes are mixed together. For the DSAN method, there is a serious misclassification between the "grass stressed" and "residential buildings" classes. For the TSTnet method, the "residential buildings" and "non-residential buildings" classes suffer from misclassification. Different from the compared approaches, the CPGAN obtains a more accurate classification map, and the visual map has less noisy labels.

TABLE IV
CLASSIFICATION RESULTS (%) USING 5% SOURCE LABELS FOR THE HOUSTON2018 DATASET

Class No.	Classification Models							
	JAN	MDD	DAAN	MCC	BNM	DSAN	TSTNet	Ours
1	44.18	49.14	52.12	54.75	42.03	49.14	**66.54**	54.61
2	60.73	68.15	67.74	74.88	64.30	65.92	**81.92**	78.58
3	54.87	51.48	56.97	59.49	41.47	49.37	51.16	**62.91**
4	20.34	8.45	9.52	28.71	6.51	12.05	10.09	**33.68**
5	49.43	49.27	45.11	59.75	39.64	37.12	49.43	**64.52**
6	82.64	80.99	83.92	87.24	85.22	84.82	87.69	**90.06**
7	31.50	33.10	36.92	51.16	40.94	34.82	**62.98**	55.18
OA	67.98	67.22	70.91	76.23	70.59	70.68	77.09	**80.12**
AA	49.10	48.65	50.33	59.42	45.73	47.61	58.55	**62.79**
Kappa	49.47	48.74	51.54	61.25	50.77	49.66	63.47	**66.29**

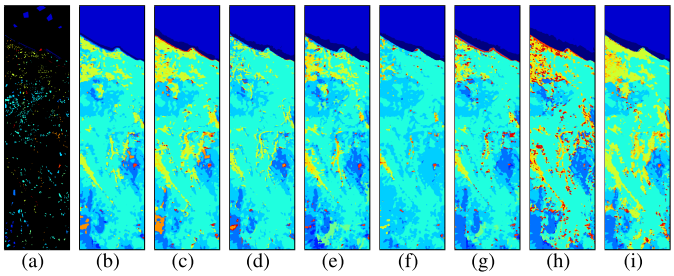
TABLE V
CLASSIFICATION RESULTS (%) USING 5% SOURCE LABELS FOR THE LOUKIA DATASET

Class No.	Classification Models							
	JAN	MDD	DAAN	MCC	BNM	DSAN	TSTNet	Ours
1	17.21	21.41	16.61	20.32	**27.96**	13.74	23.42	27.57
2	29.52	33.08	4.18	**57.68**	20.37	18.26	0.00	41.08
3	11.77	29.75	13.46	25.84	24.11	16.71	14.89	**46.35**
4	8.32	**17.57**	7.15	9.48	5.87	8.41	0.00	15.14
5	47.16	46.73	48.86	44.67	42.43	37.12	**65.47**	65.35
6	11.48	18.57	7.86	**31.27**	14.41	21.39	8.45	21.93
7	67.55	66.58	69.76	69.31	65.21	67.11	**70.93**	70.48
8	64.94	64.73	64.37	62.42	60.73	57.98	63.71	**68.37**
9	29.97	27.36	35.14	**42.45**	29.73	24.98	38.93	25.98
10	8.81	0.80	2.73	**61.80**	19.88	34.01	22.34	0.33
11	99.68	99.79	96.77	98.47	98.01	96.99	98.86	**100.00**
12	**93.15**	89.86	73.05	85.99	87.46	71.77	93.04	90.94
OA	63.08	61.96	63.58	64.96	61.34	60.29	64.73	**67.31**
AA	40.80	43.02	36.66	50.81	41.34	39.04	41.67	**47.79**
Kappa	54.12	53.50	54.31	57.09	52.55	50.78	56.89	**59.72**

Besides, Table IV presents the objective classification results of experiment methods. The MDD and BNM methods yield unsatisfactory classification accuracies. The TSTnet method boosts the classification performance. However, the fourth class cannot be well detected. Compared to other approaches, the CPGAN obtains the highest OA, AA, and Kappa. Besides, the CPGAN produces the best performance for four classes.

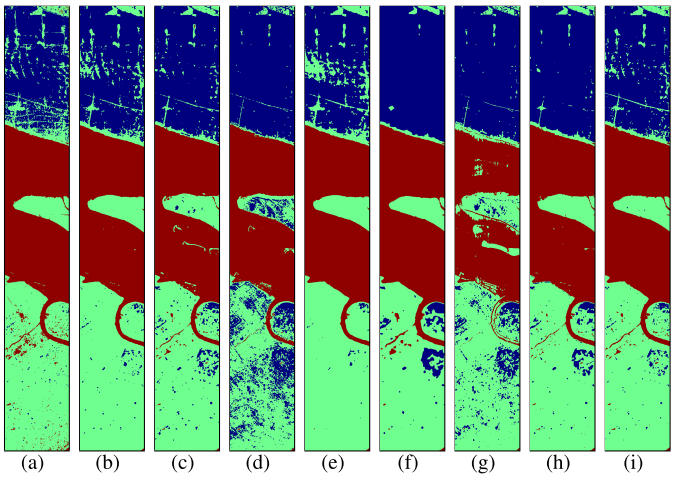
2) Hyrank: The second experiment is conducted on the Hyrank dataset. Table V lists the objective classification results of all approaches; 5% samples are selected from GT as the training set. This dataset comprises various land covers that share similarities, including Fruit Trees, Olive Groves, and Dense Sderophyllous Vegetation. The DSAN method

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 6. Classification maps of the Hyrank task. (a) GT. (b) JAN, OA = 63.08%. (c) MDD, OA = 61.96%. (d) DAAN, OA = 63.58%. (e) MCC, OA = 64.96%. (f) BNM, OA = 61.34%. (g) DSAN, OA = 60.29%. (h) TSTnet, OA = 64.73%. (i) Our method, OA = 67.31%.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 7. Classification maps of the S-H task. (a) GT. (b) JAN, OA = 88.13%. (b) MDD, OA = 88.91%. (d) DAAN, OA = 82.48%. (e) MCC, OA = 91.83%. (f) BNM, OA = 90.99%. (g) DSAN, OA = 85.47%. (h) TSTnet, OA = 91.50%. (i) Our method, OA = 94.09%.

achieves the lowest classification performance across three objective indexes. The TSTnet fails to identify the "mineral extraction sites" and "fruit trees" classes. The MCC method greatly enhances the classification performance. In general, the CPGAN obtains the highest classification accuracy. Moreover, our method obtains the highest individual classification result.

Fig. 6 shows the visual maps of all classification techniques. For the JAN method, the "coastal water" class fails to be identified well. The DAAN method suffers from serious misclassification for the "olive groves" class. The MCC method improves the visual map. However, the "olive groves" class cannot be well identified. The BNM method struggles to effectively classify the "sparse vegetated areas" class in the visual map. For the DSAN method, the "coastal water" class appears misclassification. The TSTnet method yields an oversmoothed result for the "dense urban fabric" class. In contrast, the CPGAN provides better visual classification results, which is more similar to the GT.

3) S-H: The third experiment is tested on the S-H dataset. The size of the training set is 5%. Fig. 7 presents the classification maps of all approaches. For the JAN method, the small water regions in the "building/ground" class are not identified. For the MDD method, the "plant class" is misclassified into the "building/ground" class. The DAAN method yields a serious misclassification map. The MCC method improves

TABLE VI

CLASSIFICATION RESULTS (%) USING 5% SOURCE LABELS FOR THE SHANGHAI DATASET

Class No.	Classification models							
	JAN	MDD	DAAN	MCC	BNM	DSAN	TSTNet	Ours
1	94.09	95.71	93.48	96.45	96.89	93.92	92.71	**97.50**
2	85.79	86.50	77.43	90.32	89.11	81.94	89.83	**93.22**
3	84.73	84.44	77.21	88.45	86.57	81.01	87.00	**90.94**
OA	88.13	88.91	82.48	91.83	90.99	85.47	91.50	**94.09**
AA	88.20	88.89	82.71	91.74	90.86	85.62	91.31	**93.89**
Kappa	81.94	83.17	73.81	87.53	86.28	78.11	87.02	**90.86**

the classification map. However, some "water" areas fail to be identified. The BNM and DSAN methods appear similar results in the classification maps. A large number of "plant" classes are classified into the "building/ground" class. For the TSTnet method, some minor "building/ground" regions cannot be well identified. By contrast, the classification map of the CPGAN can well align with the GT. This experiment further illustrates the advantage of the CPGAN.

Furthermore, Table VI displays the classification accuracies of all approaches. It is obvious that the CPGAN achieves the highest classification accuracies across three objective indexes. The OA is increased by 5.96% compared to the JAN method.

D. Ablation Study

In this section, to further demonstrate the effectiveness of the proposed CPGA, the influence of the proposed method with different components on the classification performance is discussed, and the results for each dataset are presented in Table VII. In this scenario, source only indicates that only source data are utilized to train the classification model; UPS denotes that the uncertainty estimation is introduced to generate pseudolabels for learning target scene information; InterDA stands for aligning only the interdomain distribution to reduce the discrepancy across scenes; ProC is the prototype calibration operation to achieve discriminative classwise prototypes, which efficiently guides the alignment of the features from different scenes; and IntraDA represents the operation of reducing the distance of intraclass features. From Table VII, it is evident that introducing uncertainty estimation is useful for learning target information compared to using only source data. In addition, aligning both the interdomain and intradomain distribution is efficient in improving the cross-scene classification accuracy. Moreover, calibrating the classwise prototypes aids the classification model in obtaining more representative prototypes to accurately guide the feature alignment. Furthermore, when all four components are considered together, the proposed method achieves the best classification performance. This demonstrates that each of the four components plays a corresponding role and contributes to the overall classification performance.

To sum up, InterDA and IntraDA can effectively reduce feature discrepancy across scenes. The UPS can select high-quality pseudolabels to help the model learn target knowledge. In addition, the ProC can enhance the accuracy of prototype representations. As shown in Table VII, it also demonstrates that each of the four components plays a

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TABLE VII
ABLATION RESULTS (%) OF THE PROPOSED METHOD WITH DIFFERENT COMPONENTS

Stages	Component				Dataset								
					Houston	Hyrank	S-H	Houston	Hyrank	S-H	Houston	Hyrank	S-H
					OA			AA			Kappa		
stage 1	Source only				71.55	55.16	89.06	50.28	37.30	88.92	51.03	45.23	83.39
stage 2	UPS	InterDA	ProC	IntraDA									
	✓				77.89	64.96	92.50	59.37	43.79	92.37	62.00	56.69	88.51
	✓	✓			78.91	65.06	92.80	61.2	43.36	92.67	63.22	56.56	88.96
	✓	✓	✓		79.58	67.11	93.25	61.25	47.77	93.13	65.28	59.47	90.52
	✓	✓	✓	✓	80.12	67.31	94.09	62.79	47.79	93.89	66.29	59.72	90.86

TABLE VIII
EXECUTION TIME OF DIFFERENT METHODS (IN SECONDS) FOR ONE EPOCH TRAINING

Datasets	JAN	MDD	DAAN	MCC	BNM	DSAN	TSTNet	Ours
Houston	1.31	**1.23**	1.73	1.24	1.34	1.46	8.01	1.27
Hyrank	10.19	9.72	14.11	9.11	10.19	**7.65**	23.71	10.89
S-H	13.74	**12.83**	16.32	13.24	15.33	15.12	28.51	14.15

corresponding role and contributes to the overall classification performance.

E. Discussion of Computing Time

Table VIII lists the computing time of all methods for each epoch. The number of training samples is set as in Tables IV–VI. All techniques are implemented with a computer with NVIDIA GTX 2080 Ti GPU. It is shown that the MDD method performs the fastest among all classification techniques. The reason is that MDD only aligns with the margin distribution. The calculation speed of the CPGAN ranks third among all approaches. For the Houston dataset, the running time of the proposed CPGAN is 1.27 s; it is still acceptable.

In summary, compared to the comparison methods such as MDD and BNM methods, the proposed CPGAN achieves an average improvement of 4%–10% in classification accuracy with minimal computational cost. In the future, we will further optimize the network structure to enhance computational efficiency while maintaining high classification accuracy.

F. Discussion of Learning Rate

To analyze the impact of different learning rates on the classification performance for each experimental dataset, we modify the learning rate from 0.1 to 0.9 while keeping all other parameters constant and obtain the classification performance for different learning rates on different datasets. As shown in Fig. 8, different learning rates result in varying classification performances, but the overall differences are not significant. Taking the results on the S-H dataset as an example, the classification accuracy remains relatively stable between 93% and 94% as the learning rate changes, indicating good classification model stability. This is because the proposed class-wise prototype-guided alignment is a more stable approach,

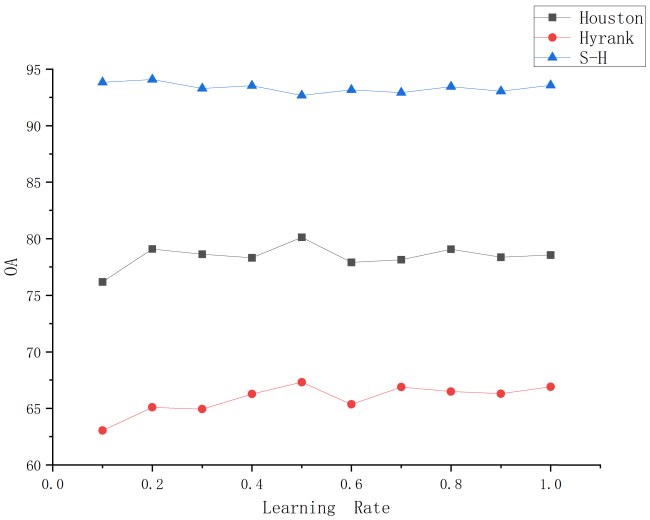


Fig. 8. Classification accuracy of different learning rates on Houston, Hyrank, and S-H datasets.

making the model less sensitive to changes in learning rates. Furthermore, for the Houston and Hyrank datasets, it is clear that the classifier achieves the best performance when the learning rate is set to 0.5. When the learning rate is less than 0.5, the classification accuracy gradually increases as the learning rate increases. In addition, the S-H dataset achieves the best classification performance with a learning rate of 0.2, while Houston and Hyrank achieve the highest classification accuracy with a learning rate of 0.5. This difference is due to the fact that compared to the Houston and Hyrank datasets, the S-H dataset has more distinct land cover distributions, only requiring a smaller learning rate to accurately classify the target scenes.

G. Discussion of Batch Size

To verify the impact of different batch sizes on the model's classification performance, we conduct experiments on the Houston dataset using various batch sizes. As shown in Fig. 9, when the batch size is set to 30, the prototype's representation capability is insufficient due to the limited number of samples used for prototype calculation, resulting in the lowest classification accuracy. As the batch size increases to 100, the classification performance of the proposed method

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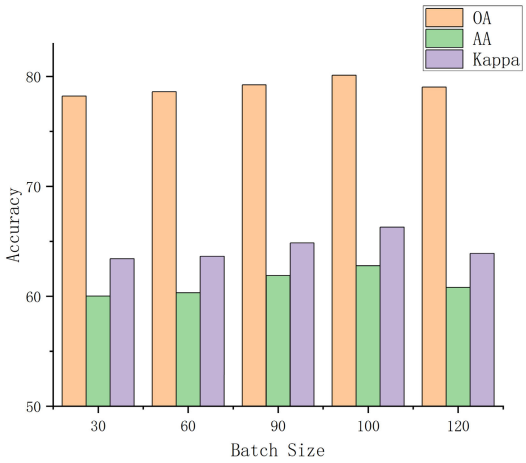


Fig. 9. Classification results of different batch sizes on the Houston datasets.

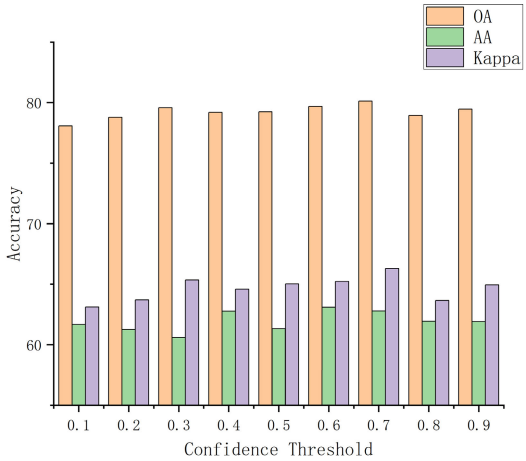


Fig. 10. Classification results of different confidence thresholds on the Houston datasets.

gradually improves. This improvement is attributed to the enhancement of the prototype's representation capability with the increasing number of samples, leading to better classification performance. However, when the batch size is set to 120, the classification accuracy decreases again. This decline is due to the introduction of a large amount of redundant information, which diminishes the model's ability to recognize important features, thereby affecting the classification accuracy of the target scenes. In summary, setting an optimal batch size is crucial for obtaining high-quality prototypes. Therefore, in this work, the batch size is set to 100 to accurately calculate the prototypes for each class.

H. Discussion of Confidence Threshold

As shown in Fig. 10, the classification accuracy on the Houston dataset varies with different τ's. The proposed CPGAN achieves the highest classification accuracy when τ is set to 0.7. However, if the threshold is set above 0.7, a high threshold will introduce a significant amount of noise, causing the model to potentially learn incorrect features and, thus, reducing its classification performance. Conversely, when the threshold is set too low, the number of pseudolabels is insufficient, leading to inadequate training samples. This prevents the model from learning the feature distribution of the target scene, resulting in lower classification accuracy. Therefore, in this study, the confidence threshold is set to 0.7 to help the model select high-quality pseudolabels.

IV. Conclusion

In this work, a CPGAN is proposed for cross-scene HSIC, which is efficient in aligning cross-scene features and enhances the classification accuracy of the target scene. In particular, we utilize ResNet 50 to extract cross-scene features at first. Second, the CPGA is proposed to align the interdomain and intradomain distribution. It is noteworthy that the ProC is applied to calibrate classwise prototypes, enhancing the stability of the prototype. Finally, the UPS is employed to obtain high-confidence pseudolabels for the target scene. Comprehensive experiments are conducted to prove the effectiveness of the proposed CPGAN.

To sum up, the primary contribution of this article is that an efficient classwise prototype network is proposed, thus achieving superior classification performance on the three datasets. However, compared with other classification methods, although the proposed CPGAN achieves the highest classification accuracy in three public datasets, CPGAN has a slightly longer calculation time. Therefore, in future research, we will further optimize the network structure and design a more lightweight alignment network to effectively reduce computation time while ensuring model accuracy.

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